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College of Informatics and Computing Sciences

"Lab Equipment Borrowing and Tracking System"

In Partial Fulfilment of the Requirement in
IT 312 - Systems Integration and Architecture

By:

GROUP 2

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Title of the Project

Lab Equipment Borrowing and Tracking System

Description / Overview

The Lab Equipment Borrowing and Tracking System is designed to provide an efficient and reliable system for managing and monitoring laboratory equipment within the school. By integrating RFID technology, the system automates the borrowing and returning process, reducing manual errors and saving time for both students and staff. Each piece of equipment will be tagged with an RFID sticker, and users will utilize their RFID-enabled IDs to borrow or return items. This eliminates the need for manual entry, ensures accurate records, and enhances accountability in the use of school laboratory resources. The system ultimately aims to streamline equipment management, promote proper utilization, and prevent loss or misplacement of laboratory assets.

General Objectives

- To develop a digital system that automates the borrowing and returning of laboratory equipment using RFID technology.
- To establish a reliable tracking mechanism that records the real-time status and history of borrowed equipment.
- To minimize manual processes and human error in logging laboratory equipment transactions.
- To promote accountability among students and staff in handling laboratory equipment.
- To support laboratory administrators in managing inventory through accurate and accessible records.

Integration Analysis

The system integrates three major components:

- Hardware Integration - Uses RFID/IoT tags, sensors, and microcontrollers to track users and equipment.
- Database Server - Centralized database stores borrow/return logs and equipment status.
- Network Integration – Devices transmit data securely to the central system.

Data flow is analyzed to ensure seamless communication between the hardware and software layers. Integration testing is performed to verify interoperability.

Agile Methodology

Sprint Planning:

The system development is divided into multiple sprints, each focusing on specific core features.

- **Sprint 1:** Implementation of user authentication through RFID-based login and sign-up features.
- **Sprint 2:** Development of the landing page and inventory management interface.
- **Sprint 3:** Integration of the borrowing and returning process with real-time equipment tracking using RFID.
- **Sprint 4:** Creation of the dashboard and report generation modules.



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- Daily Stand-ups:** Short daily check-ins to share progress, identify challenges, and set immediate goals for the next development tasks.
- Sprint Review:** Demonstration of completed features such as RFID authentication, equipment tracking functionality, database updates, and dashboard analytics to the instructor or client for feedback
- Sprint Retrospective:** The team reviews the sprint process, discusses challenges encountered, and identifies actionable ways to improve teamwork and efficiency for the next cycle.

User and System Requirements (Hardware and Software)

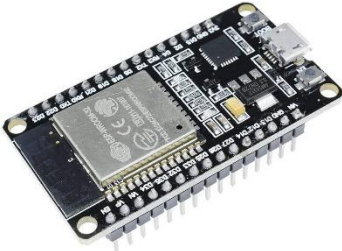
For User:

- Must be able to borrow and return laboratory equipment quickly and easily.
- Must have a unique RFID ID for identification during transactions.
- Should be able to check their borrowing history and status.

Laboratory Staff:

- Must be able to scan equipment to record borrowing and returning.
- Should be able to verify the borrower’s identity via RFID.
- Must have access to a dashboard to view current availability of items.
- Should be able to generate reports on borrowed, returned, and missing equipment.

For Hardware Requirements:

HARDWARE	DEVICE ILLUSTRATION	FUNCTION
Microcontroller: ESP32		Serves as the main processing unit, handling RFID scans and communicating with the database/server through Wi-Fi.

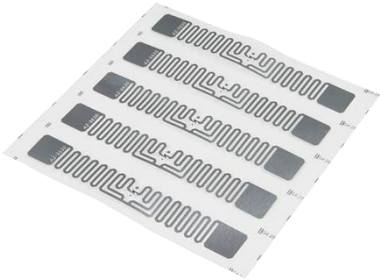
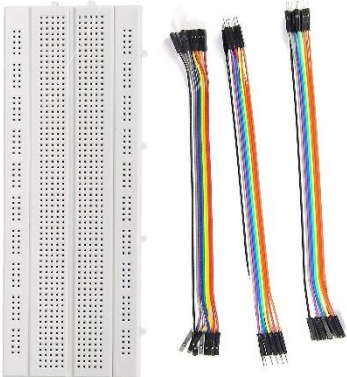


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RFID Reader: RC522 Module		Used for scanning RFID cards/tags assigned to users and equipment.
RFID Tags/Stickers		Each tag contains a unique ID used for identifying students and equipment during borrowing or returning.
LED Indicators		Provide visual feedback (e.g., green for success, red for error).
Breadboard & Jumper Wires		Used for circuit prototyping and component connections.
Power Supply/USB Cable		Provides power to the ESP32 and peripheral components during testing and operation.



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For Software Requirements:

SOFTWARE	DESCRIPTION
Operating System	Windows
Programming Language	Java / C++
Database	MySQL or PostgreSQL
Development Environment	Visual Studio Code / Eclipse IDE
Firmware	Arduino IDE or PlatformIO

System Model / Design Diagram

Description:

The system follows a **3-Tier Architecture** adapted for an application setup:

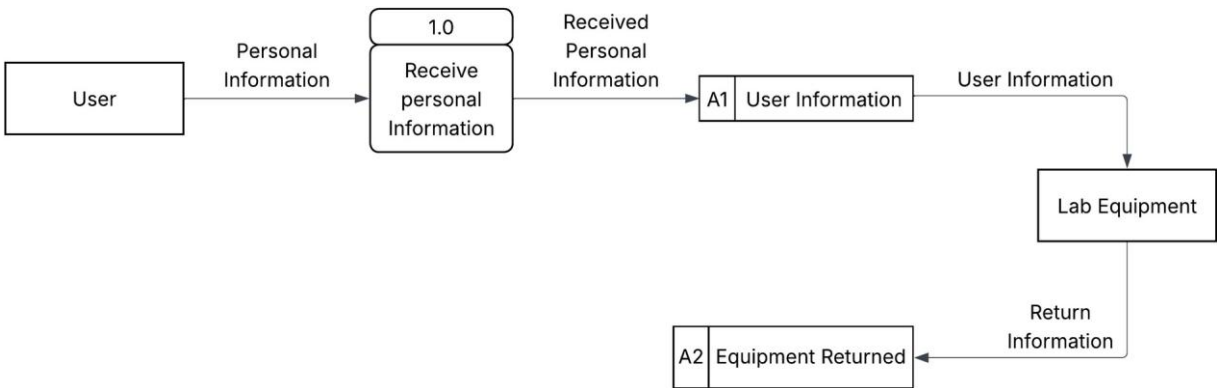
Presentation Layer – Provides the interface for students and administrators to borrow/return items, track equipment status, and view logs.

Application Layer – Processes input from the RFID hardware, verifies transactions and enforces/returning rules.

Data Layer – Centralized database storing student information, equipment inventory, and transaction logs.

Data Flow Diagram

Borrowing Process of Laboratory Equipment



1. Receive Personal Information (Process 1.0)

The system collects and validates the user’s personal information. This information is then transmitted to the **User Information (A1)** data store for processing and verification.

2. User Information and Lab Equipment Integration

After verification, the user’s information is linked with the corresponding Lab Equipment that the user intends to borrow. The system retrieves the relevant equipment data and associates it with the borrowing user.



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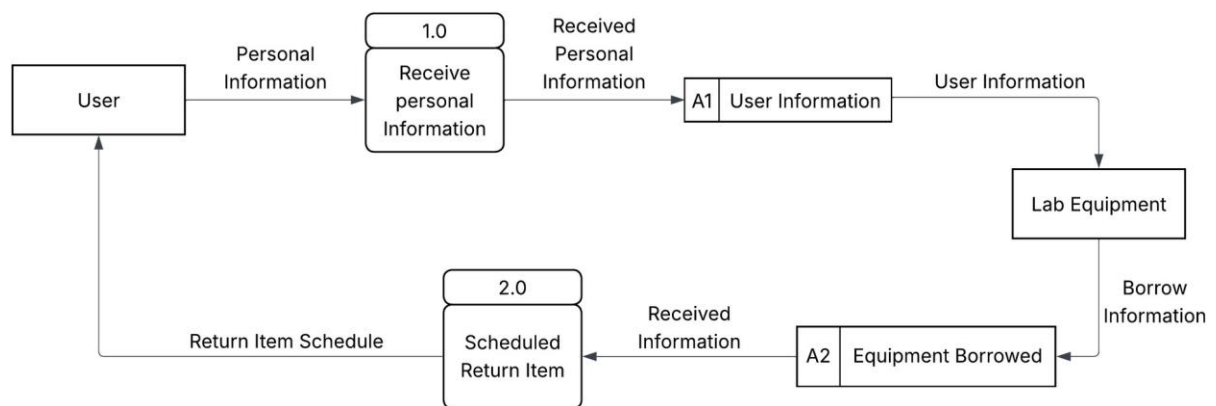
3. Equipment Borrowed (A2)

The borrowing details are stored in the **Equipment Borrowed** data store. The system subsequently updates the database to reflect the change in equipment status from “*Available*” to “*Borrowed*.”

4. Scheduled Return Item (Process 2.0)

The user specifies the intended return date for the borrowed equipment. The system records the **return schedule** and updates the transaction record accordingly, completing the borrowing process.

Returning Process of Laboratory Equipment



1. Receive Personal Information (Process 1.0)

The system receives and verifies the user’s information to identify which items were previously borrowed under that user’s account.

2. User Information and Lab Equipment Interaction

The verified user data is matched with the corresponding **Lab Equipment** that is being returned.

3. Equipment Returned (A2)

Once the return transaction is confirmed, the system records the event in the **Equipment Returned** data store. The database is then updated to change the equipment status from “*Borrowed*” to “*Available*.”